Homologous Series

Exam code: 0620 & 0971

Download PDF

Written by: Alexandra Brennan
Reviewed by: Stewart Hird
Updated on 16 December 2024

Homologous series

- A homologous series is a family of organic compounds that have similar features and chemical properties due to them having the same **functional group**
- The functional group is a group of atoms which are bonded in a specific arrangement that is responsible for the characteristic reactions of each member of a homologous series

Table of compounds & their functional groups

Family	Functional Group	Name ends in...
Alkane	C-C	-ane
Alkene	C=C	-ene
Alcohol	-OH	-ol
Carboxylic acid	-COOH	-anoic acid
Ester	-COO-	-yl-anoate

Examiner Tips and Tricks

Make sure you can identify the functional group for each homologous series.

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

General formulae

- The **general formula** tells you the composition of any member of a whole homologous series of organic compound
- For example:

are in the alkane, doubling this number and adding two will give you the number of hydrogen atoms present in the alkane

- General formulae can be used to work out the formula of a compound from different homologous series if the number of carbon atoms present is known

General formula of common homologous series

Homologous Series	General Formula
Alkane	C_nH_{2n+2}
Alkene	C_nH_{2n}
Alcohol	$C_nH_{2n+1}OH$
Carboxylic acid	$C_nH_{2n+1}COOH$

Worked Example

Questions

1. What is the formula of an alcohol that contains 5 carbon atoms?
2. What is the formula of an alkene that contains 10 carbon atoms?

Answers:

1. The formula of an alcohol containing 5 carbon atoms is:

- Number of carbons = 5
- Number of hydrogen atoms (excluding in the functional group) = 2 x 5 + 1 = 11
- Formula = $C_5H_{11}OH$

2. The formula of an alkene that contains 10 carbon atoms is:

- Number of carbons = 10
- Number of hydrogen atoms = 2 x 10 = 20
- Formula = $C_{10}H_{20}$

General characteristic of homologous series

Extended tier only

- All members of a homologous series have:
  - The **same general formula**
  - **Same functional group**

- The difference in the molecular formula between one member and the next is CH_2
- These characteristics are shown below for ethanol and propanol, which belong to homologous series, alcohols

Table of characteristics of ethanol and propanol

	Ethanol	Propanol
Molecular formula	CH_3CH_2OH	$CH_3CH_2CH_2OH$
General formula	$C_nH_{2n+1}OH$	$C_nH_{2n+1}OH$
Functional group	-OH	-OH
Boiling point (°C)	78	97

Examiner Tips and Tricks

Make sure you learn the general formula for each homologous series.

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder

Test yourself

Flashcards

Was this revision note helpful? Yes No

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.
Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.